

Statistical and Machine Learning

Introduction

Instructions

All questions should first be discussed in pairs or small groups. Afterwards, we shall discuss them together.

Question 1

- What is machine learning (ML)?
 - Explain ML to a 10-year-old
 - Explain ML to an undergraduate Chemistry student
 - Your colleague studying Physics says that statistics and ML are old school, and we should all use AI to work with data. Is that true? What would you respond to him/her/them?
-

Question 2: Iris data

We consider the `iris` data set (Example 1, lecture slides), where the goal is to classify an iris flower into one of three species based on its measured features.

1. What is the input space?
 2. What is the task at hand?
 3. What optimality criterion would you use to build the learner?
 4. Should the linear or non-linear classifier be used?
-

Question 3: Image data

Consider image data from ImageNet (General image classification, lecture slides). Suppose the goal is to recognize whether a photo contains a cat.

1. What is the input space? Consider images in colour of sizes 32×32 pixels. What is the number of features, p , equal to?
 2. What is the task at hand?
 3. What optimality criterion would you use?
 4. Why is this problem more difficult than the iris example?
 5. Why might classical statistical methods have difficulty solving this problem well?
 6. Can you give an example of another problem where ML/AI perform exceptionally well, whereas traditional statistical methods struggle?
-

Question 4: Supervised, unsupervised, and semi-supervised learning

1. Define a supervised learning problem, list a few methods useful for solving this problem, and give a data example to which one can apply supervised learning.
2. Define an unsupervised learning problem, list a few methods useful for solving this problem, and give a data example to which one can apply unsupervised learning.

3. Can you think of a definition of semi-supervised learning?
-

Question 5: Real-life case study – building a learner at Amezo

Imagine you work as a data scientist at *Amezo*, a large online retailer. You are asked to build a learner to increase the sales of cleaning products.

A possible goal is to predict which customers are most likely to buy a cleaning product after seeing a recommendation or discount.

Possible predictors include:

- past purchases,
- browsing history,
- prices and discounts,
- season,
- customer demographics.

1. How would you start solving this task?
2. What could be the input space and output space?
3. What is the task at hand?
4. What optimality criterion would you use?
5. How would you build the learner?
6. How would you test whether the learner is good?
7. When can we trust and use the learner?